

Network Pharmacology Analysis of EGCG, Tanshinone I, and E-Guggulsterone Against Breast Cancer

1. Research Background and Rationale

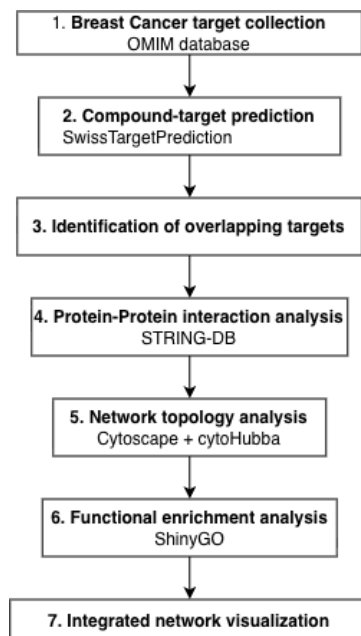
Breast cancer is a molecularly heterogeneous disease involving multiple genetic alterations and interconnected signaling pathways. Tumor cells can activate alternative survival and growth mechanisms that support proliferation, apoptosis resistance, invasion, and resistance to endocrine or targeted therapies [1]. Network pharmacology is therefore suitable for investigating breast cancer because it integrates compound targets, disease-associated genes, protein–protein interactions, and enriched biological pathways within a unified analytical framework [2].

EGCG, tanshinone I, and E-guggulsterone were selected because they represent different natural-product chemical classes and have shown anticancer activity in preclinical breast-cancer models. EGCG induced apoptosis in T47D cells through modulation of the PI3K–AKT pathway and suppression of telomerase-related expression [3]. Tanshinone I induced apoptosis in both estrogen receptor-positive MCF-7 and estrogen receptor-negative MDA-MB-231 cells [4] and suppressed tumor growth, invasion, and angiogenesis in MDA-MB-231 models [5]. E-guggulsterone corresponds to the E or trans stereoisomer of guggulsterone; in MCF-7 cells, trans-guggulsterone suppressed MAPK/AP-1 signaling, MMP-9 expression, and cellular invasion [6]. Together, these findings provide a biological basis for comparing the predicted targets and pathways of the three compounds in a single network-pharmacology analysis.

2. Research Objective

The objective of this analysis was to identify possible protein targets and biological pathways through which EGCG, Tanshinone I, and E-Guggulsterone may act against breast cancer, and to determine the central proteins that may be considered for further analysis such as molecular docking or experimental validation.

3. Methodology



4. Results and Interpretation

4.1 Core protein-protein interaction network

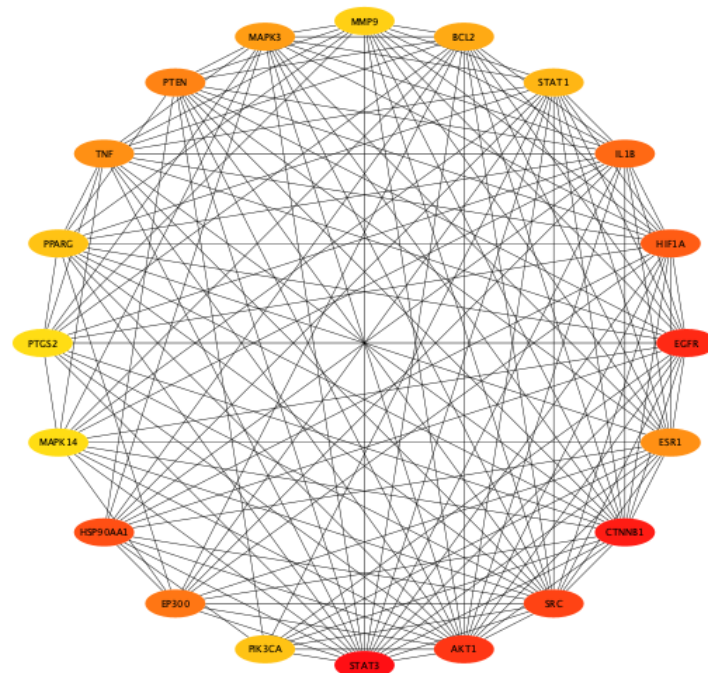


Figure 2. Core protein-protein interaction network generated from the candidate breast-cancer targets. Node color appears to represent relative topological importance, while edges show protein-protein associations.

The core network contains 20 highly connected proteins. The network is dense, meaning that many of the proteins interact with several others. This supports a multi-target interpretation rather than a mechanism controlled by one isolated protein.

Based on the visual color scale, STAT3, CTNNB1, EGFR, SRC, and AKT1 appear among the most central nodes. These proteins connect major cancer processes. EGFR, SRC, PIK3CA, AKT1, and PTEN form a growth and survival signaling group. MAPK3 and MAPK14 connect mitogen and stress-response pathways. ESR1, PPARG, and EP300 connect hormone-receptor and transcriptional regulation. TNF, IL18, STAT1, STAT3, and PTGS2 represent inflammatory or immune signaling. HIF1A and MMP9 relate to hypoxia, angiogenesis, tissue invasion, and metastasis, while BCL2 is involved in resistance to apoptosis.

4.2 Identification of Hub Genes in the Core Network

Gene	Betweenness Centrality	Closeness Centrality	Degree
AKT1	0.022219717	1.000000000	19
STAT3	0.022219717	1.000000000	19
SRC	0.021144726	0.950000000	18
CTNNB1	0.018687885	0.950000000	18
EGFR	0.016352988	0.904761905	17
ESR1	0.014972368	0.904761905	17
IL1B	0.013755883	0.904761905	17
HIF1A	0.011396509	0.904761905	17

STAT1	0.014348479	0.863636364	16
BCL2	0.010691398	0.863636364	16
MAPK3	0.008233298	0.826086957	15
MMP9	0.005026352	0.826086957	15
PTEN	0.007542046	0.791666667	14
PPARG	0.006173093	0.791666667	14
TNF	0.004044877	0.791666667	14
HSP90AA1	0.006883348	0.760000000	13
MAPK14	0.004829603	0.760000000	13
PTGS2	0.003139837	0.760000000	13
EP300	0.003812459	0.730769231	12
PIK3CA	0.000899685	0.655172414	9

The protein–protein interaction network contained 20 genes and 153 interactions, showing a highly connected network. Centrality analysis identified AKT1 and STAT3 as the main hub genes, as both had the highest degree, betweenness centrality, and closeness centrality values. SRC and CTNNB1 were also highly central, followed by EGFR, ESR1, IL1B, and HIF1A. These genes may act as important connection points between several breast cancer-related processes, including cell proliferation, survival, hormone signaling, inflammation, and response to cellular stress.

Overall, the results suggest that the possible effects of EGCG, tanshinone I, and E-guggulsterone may involve multiple interconnected signaling pathways rather than a single molecular target. AKT1, STAT3, SRC, and CTNNB1 may therefore be considered the most important candidates for further analysis. However, centrality only shows the structural importance of a gene within the selected network and does not confirm direct compound–target binding or therapeutic activity.

4.3 Gene Ontology molecular-function enrichment

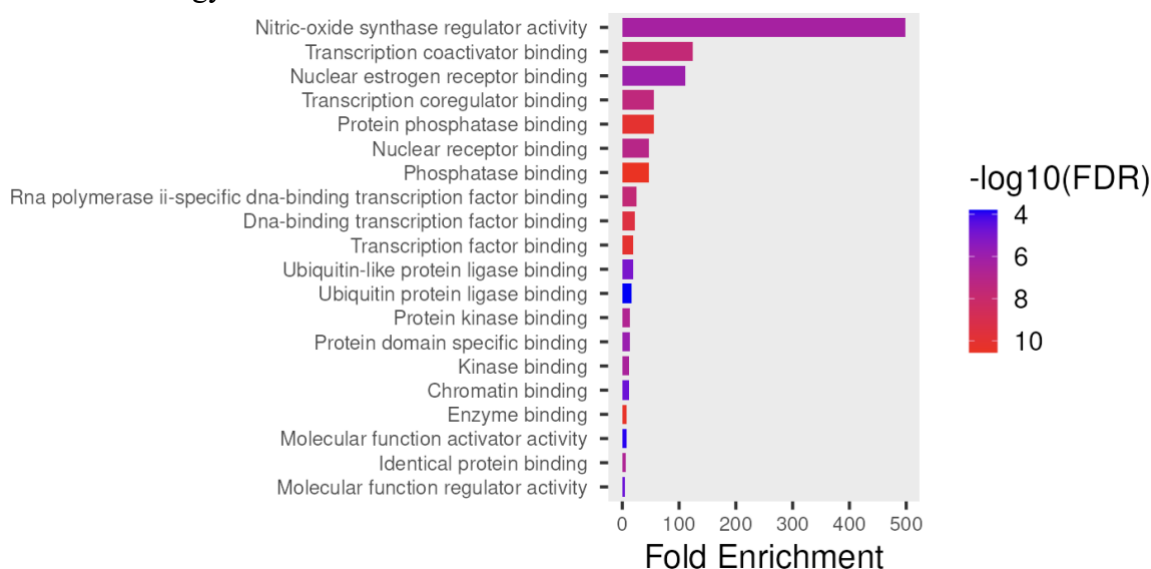


Figure 3. Gene Ontology Molecular Function enrichment.

The largest fold-enrichment value was observed for nitric-oxide synthase regulator activity. This means that this molecular function occurred much more frequently in the selected gene set than expected from the reference genome. However, an extreme fold-enrichment value can be produced by a small number of genes when the background category is small. Therefore, this term should not automatically be treated as the main mechanism without checking the number and identity of contributing genes.

The broader pattern is more informative. Enriched terms included transcription coactivator binding, nuclear estrogen receptor binding, transcription coregulator binding, nuclear receptor binding, protein phosphatase binding, protein kinase binding, chromatin binding, and ubiquitin-ligase binding. These functions match the core network, which contains ESR1, PPARG, EP300, STAT proteins, kinases, phosphatase-related regulators, and HSP90AA1. The result suggests that the candidate compounds may affect breast-cancer biology through regulation of signaling proteins and transcriptional complexes.

4.4 KEGG pathway enrichment

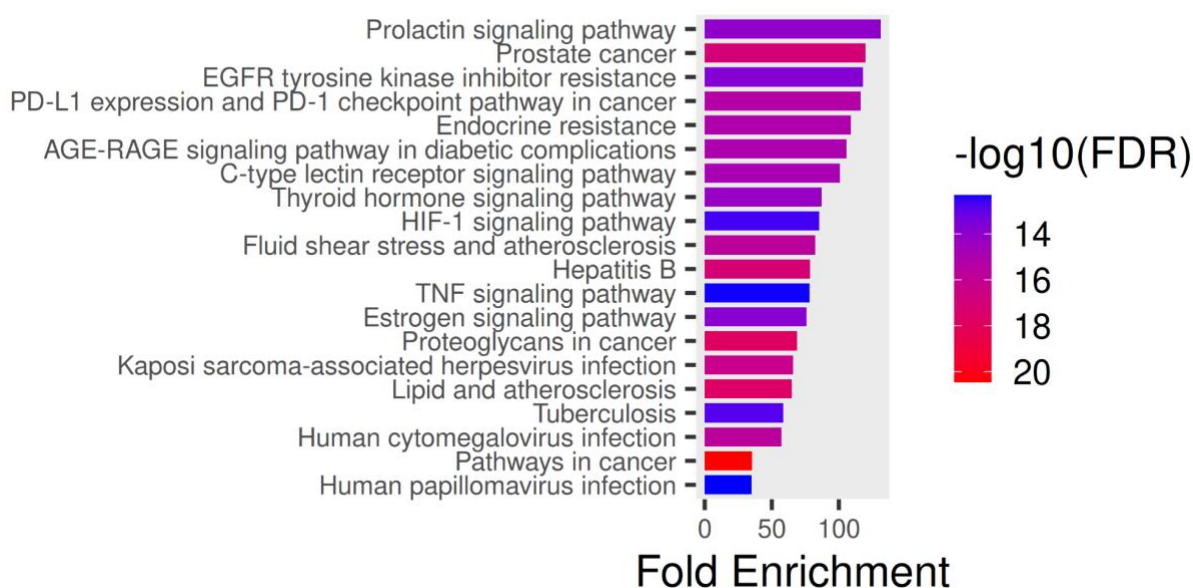


Figure 4. KEGG pathway enrichment of the candidate targets.

KEGG enrichment analysis showed that the core targets were mainly associated with interconnected pathways involved in hormonal signaling, growth-factor responses, immune regulation, hypoxia, and treatment resistance, particularly prolactin, estrogen, EGFR, HIF-1, TNF, and PD-L1/PD-1 signaling. These pathways largely converge on PI3K–AKT, MAPK, STAT, and HIF-1 signaling modules that contribute to breast-cancer progression and therapeutic response. Enrichment of prostate cancer and infection-related pathways likely reflects shared signaling proteins rather than disease-specific effects; therefore, interpretation should focus on the underlying molecular mechanisms rather than individual KEGG pathway labels.

4.5 Integrated compound-target-pathway network

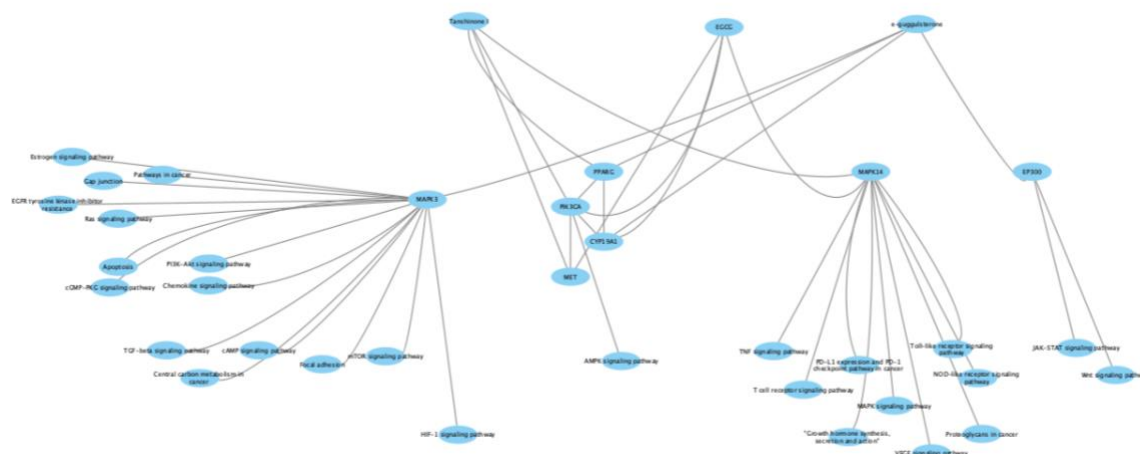


Figure 5. Integrated Cytoscape network connecting EGCG, Tanshinone I, E-Guggulsterone, selected targets, and enriched KEGG pathways.

The integrated network revealed that EGCG, tanshinone I, and E-guggulsterone converge on a limited number of highly connected targets that link the compounds to multiple cancer-related pathways. MAPK3 and MAPK14 formed important connections between growth-factor, inflammatory, stress-response, apoptosis, and immune-related pathways, whereas EP300 linked several transcriptional regulatory pathways, including JAK–STAT and Wnt signaling. PIK3CA, MET, CYP19A1, and PPARG further connected the network with cell-survival signaling, receptor tyrosine kinase activity, estrogen biosynthesis, and nuclear-receptor regulation. This network structure supports a multi-target model in which the three compounds may affect complementary but overlapping regulatory processes in breast cancer.

5. Conclusion

The network pharmacology analysis suggests that EGCG, tanshinone I, and E-guggulsterone may influence breast cancer through interconnected hormonal, growth-factor, inflammatory, hypoxia, and survival-related pathways. Among the predicted targets, AKT1, STAT3, SRC, CTNNB1, EGFR, ESR1, HIF1A, MAPK3, and MAPK14 emerged as key proteins implicated in breast-cancer pathophysiology, particularly in the regulation of cell proliferation, survival, treatment resistance, immune signaling, and tumor progression. Overall, the findings support a multi-target and multi-pathway mechanism, although experimental validation is still required to confirm the predicted biological effects.

6. References

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